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TOPIC: Machine Learning–Based Digital Agriculture Advisory System for Crop Selection in Rwanda

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CHAPTER ONE: GENERAL INTRODUCTION

1. Introduction

The Machine Learning-Based Digital Agriculture Advisory System for Crop Selection in Rwanda is a web-based system that uses Machine Learning techniques to help smallholder farmers in Rwanda choose the right crops to plant. The system uses data such as soil type, weather conditions, rainfall, altitude, and district location to generate crop recommendations. Machine Learning is used to analyze historical agricultural data and learn patterns between environmental conditions and crop performance. Based on this analysis, the system predicts the most suitable crops for a specific location.

More than 70% of people in Rwanda depend on agriculture for their livelihoods. However, farmers face many challenges such as poor soil conditions, climate change, and limited access to agricultural advice. Many farmers do not receive sufficient support from agricultural extension officers. This system helps to address this problem by providing data-driven crop recommendations using Machine Learning. The system recommends suitable crops such as maize, beans, cassava, rice, and Irish potatoes. It also provides basic planting guidance through a simple web interface that farmers can easily use.

2. Problem Statement

Many smallholder farmers in Rwanda, especially in rural areas, do not have access to proper agricultural guidance. Most farmers make farming decisions without accurate information. They plant crops that are not suitable for their soil or climate, and some farmers also use fertilizers incorrectly. This results in poor harvests and financial losses. Agricultural studies show that crop production in Rwanda is lower than expected, and one major reason is that agricultural extension officers are too few to reach all farmers [4].

Rwanda's agricultural sector faces several challenges, including limited agricultural extension services, soil degradation, climate change, lack of farming information, and incorrect crop selection. Farmers do not have access to a simple digital system that uses Machine Learning to analyze farming data and provide location-based crop recommendations. As a result, farmers experience low yields and food insecurity.

3. Background

In many developing countries, digital technology is considered an important tool for improving agriculture. International organizations such as the Food and Agriculture Organization (FAO) and the World Bank show that using data and Machine Learning in agriculture can help increase food production, improve farm efficiency, and reduce the impact of climate change [1], [2]. These technologies enable farmers to make better decisions based on real data instead of relying on experience or guesswork.

Machine Learning techniques are increasingly used in agriculture to analyze large volumes of data such as soil properties, rainfall, temperature, and historical crop yields. Studies show that Machine Learning models can accurately predict suitable crops, improve yield estimation, and support better farm planning [3]. Decision support systems that integrate Machine Learning help farmers select crops that match their local environmental conditions, which leads to higher productivity and reduced losses.

In Rwanda, the government has prioritized agricultural modernization through programs such as the Strategic Plan for Agriculture Transformation (SPAT). These initiatives have improved farmers' access to agricultural inputs, extension services, and markets [4]. Despite these efforts, many farmers still lack timely and personalized agricultural advice. Agricultural extension workers are responsible for supporting a large number of farmers, making it difficult to provide location-specific and data-driven guidance.

Most existing digital agriculture systems in Rwanda provide general farming information and are not designed to fully consider local farming conditions. Many systems do not apply Machine Learning to analyze variations in soil types, rainfall patterns, altitude, and agro-ecological zones across the country. Platforms such as the Rwanda Agricultural Board's e-extension services offer useful information but lack predictive capabilities. Other systems such as Esoko and Digital Green mainly focus on market information and do not use Machine Learning models to generate crop recommendations tailored to individual farm conditions [5]. Therefore, there is a strong need for a Machine Learning-based Digital Agriculture Advisory System that can provide location-specific crop recommendations tailored to Rwanda's agricultural environment.

4. Objectives

4.1 Main Objective

To develop a web-based Machine Learning-Based Digital Agriculture Advisory System that predicts suitable crops for smallholder farmers in Rwanda using historical and current agricultural data.

4.2 Specific Objectives

i. To collect and analyze historical agricultural data, including soil types, rainfall patterns, and crop performance in Rwanda.

ii. To integrate current weather and environmental data into the system for timely advisory services.

iii. To design and develop an online platform that provides location-based crop and fertilizer recommendations.

iv. To implement a Machine Learning prediction engine that uses past and present data to support informed farming decisions.

v. To evaluate the effectiveness of the Machine Learning model in improving decision-making among smallholder farmers.

5. Research Questions

1. How can Machine Learning be used to analyze soil, weather, and location data to recommend suitable crops in Rwanda?
2. How effective is a Machine Learning-based system in improving crop selection decisions among smallholder farmers?
3. How accurate are crop recommendations generated using historical and current agricultural data?
4. What challenges affect the performance of a Machine Learning-based agriculture advisory system in Rwanda?

6. Scope and Limitations

6.1 Scope

The project focuses on developing a web-based platform accessible via desktop and mobile browsers. It covers five major crops: maize, beans, cassava, rice, and Irish potatoes. The system covers all 30 districts of Rwanda. It uses Machine Learning models trained on historical and current agricultural data to generate crop recommendations. Weather and soil data are obtained from publicly available sources. The system includes user authentication and recommendation history.

6.2 Limitations

The system relies on publicly available and simulated data due to limited real field trial data. Machine Learning model accuracy depends on the quality and completeness of the data. Internet connectivity is required to access the web system. The system provides advisory support only and does not guarantee farming outcomes or yields.

7. Geographical Scope

The project will be implemented in Rwanda, covering all 30 districts across the four provinces and the City of Kigali. Machine Learning models will be trained using data relevant to Rwandan agricultural conditions.

8. Interest of the Project

8.1 Personal Interest

This project allows the application of Information Technology and Machine Learning skills to solve real agricultural challenges. It provides practical experience in developing a data-driven web-based advisory system and supports career development in data science and agricultural informatics.

8.2 Common Interest

The project benefits farmers, agricultural extension officers, and policymakers. Farmers receive data-based crop recommendations, extension officers can reach more farmers efficiently, and the project supports Rwanda's goal of modernizing agriculture through digital technologies.

9. Organization of the Research Project

Chapter 1: General Introduction - This chapter contains the general introduction, background, problem statement, objectives, scope, and project timeline.

Chapter 2: Literature Review - This chapter discusses what other researchers have done and justifies the use of Machine Learning as the solution approach.

Chapter 3: Research Methodology and Development Methodology - This chapter presents the data collection approach, the research design, and the software development methodology used to build the system.

Chapter 4: System Analysis and Design - This chapter presents the system architecture, database design, and interface wireframes.

Chapter 5: Implementation - This chapter explains the implementation of the system based on the design.

Chapter 6: Conclusion and Recommendation - This chapter presents the main findings and recommendations for future work.

10. Project Timeline

The table below presents the planned schedule for project activities from February 2026 to June 2026, aligned with the final defense in June 2026.

Activity / Phase	February	March	April	May	June
Proposal & Planning					
Data Collection & Analysis					
System Design					
ML Model Development					
System Implementation					
Testing & Validation					
Report Writing & Documentation					
Final Defense Preparation					

Figure 1.1: Project Timeline Gantt Chart (February - June 2026)

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

The literature review aims to provide a comprehensive understanding of the current state of Machine Learning applications in agriculture, crop recommendation systems, and digital advisory tools globally and specifically in Rwanda. By examining existing studies, reports, and related works, it provides a complete list of published and unpublished studies related to the subject. This chapter also justifies the choice of Machine Learning as the core technology for the proposed system.

2.2 Theoretical Concept

The Food and Agriculture Organization of the United Nations defines precision agriculture as the use of data and technology to optimize farming decisions at the field level, enabling farmers to apply the right inputs at the right place and at the right time [1]. Machine Learning is a subset of artificial intelligence that enables computer systems to learn from data and improve their predictions over time without being explicitly programmed. In agriculture, Machine Learning models are trained on datasets containing soil properties, climate variables, and historical crop yields to predict the most suitable crops for a given location [6].

Studies show that crop selection is one of the most critical decisions a farmer makes each season, and poor crop-soil-climate matching is a leading cause of low yields in sub-Saharan Africa [3]. Traditional advisory approaches rely on static recommendation tables prepared by agronomists, which cannot adapt to the variability of individual farm conditions. Machine Learning models such as Random Forest, Decision Tree, and Naive Bayes classifiers have demonstrated strong performance in crop recommendation tasks because they can capture non-linear relationships between multiple input variables and produce location-specific predictions [9].

In Rwanda, the Rwanda Agriculture and Animal Resources Development Board (RAB) and the Ministry of Agriculture and Animal Resources (MINAGRI) have identified inadequate access to personalized agricultural advisory services as a major constraint on smallholder productivity [4]. The Strategic Plan for Agriculture Transformation Phase IV (PSTA IV) specifically calls for the use of digital technologies to scale advisory services to all farmers across Rwanda's 30 districts [4]. These policy frameworks provide the national context within which this project is positioned.

2.3 Related Work

Sharma et al. [9] conducted a comprehensive review of Machine Learning applications in precision agriculture and found that classification algorithms, particularly Random Forest and Support Vector Machines, achieved accuracies above 90% in crop suitability prediction tasks using soil and climate data. Their work demonstrates that ensemble methods are well suited for handling the multivariate nature of agricultural datasets. However, their review was conducted across diverse global contexts and did not address the specific agro-ecological conditions of Rwanda, leaving a gap that this project addresses.

Kamilaris, Kartakoullis, and Prenafeta-Boldu [3] reviewed the application of big data analytics in agriculture and found that integrating multiple data sources, including soil sensors, satellite imagery, and meteorological stations, significantly improves prediction accuracy in crop planning systems. They highlighted that the lack of localized, high-quality datasets remains the primary barrier to deploying such systems in developing countries. This finding directly informs the data collection strategy of the proposed system, which draws on RAB soil surveys, NISR seasonal reports, and Meteo Rwanda climate records.

Liakos et al. [6] surveyed Machine Learning in agriculture across four domains: crop management, livestock management, water management, and soil management. Their survey confirmed that Decision Tree and Random Forest models are the most widely used for crop recommendation due to their interpretability and robustness to noisy data. They also noted that web-based deployment of such models is limited in sub-Saharan Africa, underscoring the practical contribution of the proposed system.

Tsan et al. [5] analyzed the digitalization of African agriculture and found that while numerous digital agriculture platforms exist across Africa, most provide static information and lack predictive capabilities based on Machine Learning. They noted that platforms such as Esoko and Digital Green focus primarily on market price information and do not generate personalized crop recommendations, confirming the need for a Machine Learning-driven system tailored to Rwandan farming conditions.

The World Bank [2] emphasized that data-driven advisory systems can increase smallholder yields by 10 to 20 percent when recommendations are personalized to local conditions. The report highlighted that digital advisory systems are most effective when accessible through

mobile and web interfaces, a consideration that directly shapes the interface design of the proposed system.

2.4 Proposed System

The proposed system is a web-based Machine Learning-Based Digital Agriculture Advisory System for Crop Selection in Rwanda. The system will collect soil, climate, and geographic data for a given district and apply trained Machine Learning models to recommend the most suitable crops for planting. The system aims to improve crop selection decisions among smallholder farmers by providing accessible, data-driven recommendations through a simple web interface.

2.4.1 Requirements Analysis

2.4.1.1 Functional Requirements

1. The system shall collect and pre-process soil, climate, and geographic data for all 30 districts of Rwanda.
2. The system shall apply a trained Machine Learning model to generate crop recommendations based on user-provided farm data.
3. The system shall display recommended crops along with basic planting guidance for each recommendation.
4. The system shall allow users to register, log in securely, and view their recommendation history.
5. The system shall integrate real-time weather data through an external API to support timely advisory outputs.

2.4.1.2 Non-Functional Requirements

- ✔ The system shall be accessible via desktop and mobile web browsers without requiring software installation.
- ✔ The system shall return crop recommendations within an acceptable response time.
- ✔ The system shall be scalable to accommodate additional crops, districts, and data sources in future versions.
- ✔ The system shall maintain compatibility with existing agricultural data formats used by RAB and NISR.

- ✔ The system shall store user data securely with appropriate authentication and access controls.

2.4.1.3 System Requirements

2.4.1.3.1 Hardware Requirements

The hardware required for this project includes a computer with sufficient processing power to handle data analysis and Machine Learning model training. A stable internet connection with adequate bandwidth is required for accessing datasets, downloading software libraries, and deploying the web application.

2.4.1.3.2 Software Requirements

The operating system can be either Linux or Windows. The database system will be MySQL for storing and managing agricultural and user data. The web server will be Apache HTTP Server. The frontend will use HTML, CSS (Bootstrap), and JavaScript. The backend will be developed using Python with the Django framework. Machine Learning tasks will use Python libraries including scikit-learn and XGBoost. Data analysis will be conducted using Jupyter Notebook, and data visualization will use Matplotlib, Seaborn, and Plotly. The development environment will use Visual Studio Code or PyCharm.

2.4.2 Expected Outcome

The expected outcome of the proposed system is a functional web-based platform that accurately recommends suitable crops to smallholder farmers in Rwanda based on their soil type, climate conditions, and district location. The system is expected to improve crop selection decisions, reduce crop failures, and contribute to increased agricultural productivity across Rwanda's farming communities.

CHAPTER THREE: RESEARCH METHODOLOGY AND DEVELOPMENT METHODOLOGY

3. Research Methodology

This chapter describes the approach used to design, develop, and validate the Machine Learning-Based Digital Agriculture Advisory System for Crop Selection in Rwanda. It covers the research design, the sources and types of data used, and the software development methodology applied to build the system.

3.1 Research Design

This project follows an applied research design, which focuses on using existing knowledge and technologies to solve a specific practical problem. The problem identified is the inability of smallholder farmers in Rwanda to access timely, location-specific crop selection advice. The solution is developed by collecting agricultural data, training Machine Learning models on that data, and deploying the models within a web-based advisory platform.

The research combines quantitative data analysis, where crop yield records, soil measurements, and climate figures are processed numerically, with a system development component, where the findings of the analysis are translated into a working software application. This combination ensures that the system is both scientifically grounded and practically useful.

3.2 Data Sources

The quality and relevance of the data used to train the Machine Learning models directly determines the accuracy of the crop recommendations the system produces. The following data sources were identified and used in this project.

3.2.1 Rwanda Agriculture and Animal Resources Development Board (RAB)

RAB maintains district-level records of crop performance, soil classification maps, and seasonal planting calendars for Rwanda. These records were used to compile the training dataset for the Machine Learning model, providing information on which crops have historically performed well under specific soil and climate conditions across Rwanda's agro-ecological zones [7].

3.2.2 National Institute of Statistics of Rwanda (NISR)

The NISR publishes seasonal agricultural survey reports that document crop yields by district, covering crops including maize, beans, cassava, Irish potato, rice, and sorghum. Data from these

surveys for the period 2018 to 2023 were used to build the historical performance component of the training dataset [8].

3.2.3 Rwanda Meteorology Agency (Meteo Rwanda)

Meteo Rwanda provides historical and seasonal climate data including rainfall totals, average temperatures, and humidity levels by district. This data was used to populate the climate input variables of the Machine Learning model, ensuring that recommendations reflect actual weather patterns in each district [10].

3.2.4 SoilGrids and Open-Meteo APIs

For districts where locally measured soil data was incomplete, the SoilGrids global soil information service was used to obtain predicted values for soil pH, nitrogen, phosphorus, and potassium content. The Open-Meteo API was used to retrieve current and forecast weather data to support real-time recommendation generation within the deployed web system.

3.3 Data Pre-processing

Raw agricultural data from the sources described above required cleaning and transformation before it could be used to train Machine Learning models. Missing values in soil nutrient fields were filled using district-level averages derived from the SoilGrids API. Categorical variables such as district name and soil type were encoded numerically using label encoding. Continuous variables including rainfall, temperature, and soil pH were normalized to a common scale to prevent any single variable from dominating the model's predictions. The final processed dataset contains records for Rwanda's 30 districts covering five crop types, with each record consisting of eight input features: soil pH, nitrogen content, phosphorus content, potassium content, average annual rainfall, average temperature, average humidity, and altitude [11].

3.4 Machine Learning Model Selection

Four supervised classification algorithms were selected for training and comparison based on their established performance in crop recommendation literature: Random Forest, Decision Tree, Naive Bayes, and K-Nearest Neighbor (KNN). Each algorithm was trained on 70% of the dataset and evaluated on the remaining 30% using accuracy, precision, recall, and F1-score as performance metrics [12]. The algorithm achieving the highest F1-score on the test set was selected for integration into the web advisory system. Random Forest was anticipated to perform

best given its robustness to overfitting and its ability to handle multiple correlated input features, which is a common characteristic of agricultural datasets [6].

3.5 System Development Methodology

The Agile methodology was selected as the software development framework for this project. Agile organizes development into short, iterative cycles called sprints, each producing a working and testable component of the system. This approach is well suited to a project where requirements may be refined as the Machine Learning models are evaluated and user feedback is gathered. It also supports the parallel development of the data pipeline, model training, and web interface components without one blocking the other.

3.5.1 Agile Methodology Diagram

The diagram below illustrates the six sprints of the Agile development cycle applied in this project, showing how each phase connects to the central iterative process.

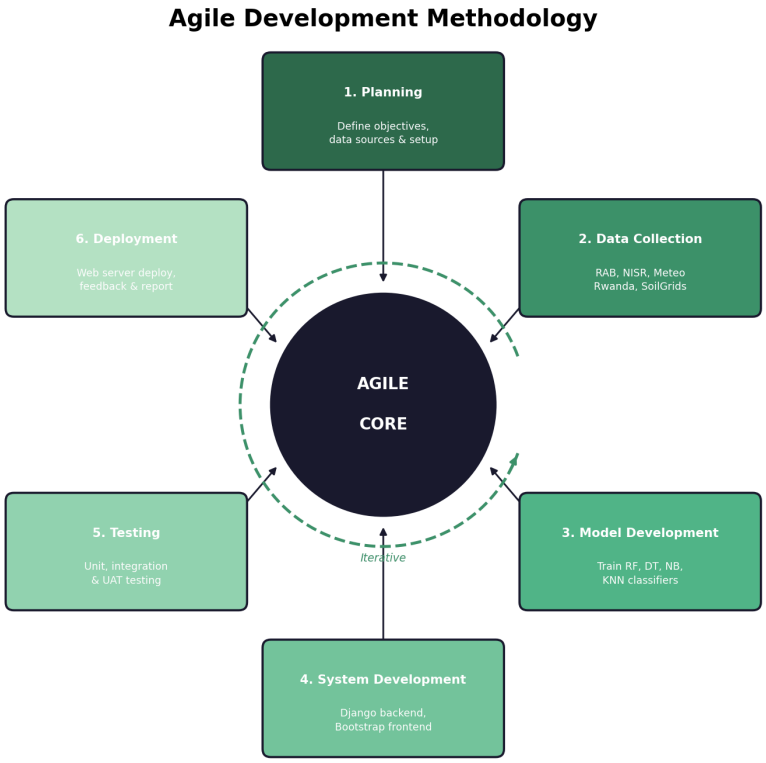

Figure 3.1: Agile Development Methodology for the Agriculture Advisory System

3.5.2 Agile Sprint Descriptions

Sprint 1 — Planning: All data source agreements were finalized, version control was configured using Git, the development environment was set up with Python, Django, and MySQL, and the complete system requirements were documented.

Sprint 2 — Data Collection: Datasets were downloaded and merged from RAB, NISR, Meteo Rwanda, and the SoilGrids API. Data cleaning, encoding, and normalization were applied to produce a ready-to-train dataset.

Sprint 3 — Model Development: The four selected Machine Learning classifiers were trained and evaluated. The best-performing model was selected based on F1-score and saved as a serialized file for backend integration.

Sprint 4 — System Development: The Django web application was built, the MySQL database schema was designed, the Bootstrap frontend interface was developed, and the trained model was connected to the crop recommendation endpoint.

Sprint 5 — Testing: Unit tests were performed on individual modules, integration tests covered the full request-response cycle, and user acceptance testing was conducted with a sample of farmers and extension officers.

Sprint 6 — Deployment and Review: The system was deployed to a web server, user feedback was collected, final issues were resolved, and the project report was completed.

3.5.3 Conclusion

The combination of applied research design, multi-source agricultural data collection, structured data pre-processing, comparative Machine Learning model evaluation, and Agile sprint-based development provides a rigorous and practical framework for building the proposed crop advisory system. This methodology ensures that the system is grounded in real Rwandan agricultural data, technically sound, and developed in a structured and testable manner.

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